MSG No. 131.436 $P4_2/mmc1'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: mmm.1'

No.	position	mapping
1	$[0, 0, 0]$	[1,4,5,6,9,12,13,14,17,20,21,22,25,28,29,30]
2	$\left[0, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16,18,19,23,24,26,27,31,32]

Table 2: Wyckoff site: 2b, site symmetry: mmm.1'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14,17,20,21,22,25,28,29,30]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16,18,19,23,24,26,27,31,32]

Table 3: Wyckoff site: 2c, site symmetry: mmm.1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14,17,20,21,22,25,28,29,30]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16,18,19,23,24,26,27,31,32]

Table 4: Wyckoff site: 2d, site symmetry: mmm.1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,4,5,6,9,12,13,14,17,20,21,22,25,28,29,30]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,3,7,8,10,11,15,16,18,19,23,24,26,27,31,32]

Table 5: Wyckoff site: 2e, site symmetry: -4m21'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,6,7,8,10,11,12,13,17,22,23,24,26,27,28,29]
2	$\left[0, 0, \frac{3}{4}\right]$	[2,3,4,5,9,14,15,16,18,19,20,21,25,30,31,32]

Table 6: Wyckoff site: 2f, site symmetry: -4m21'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[1,6,7,8,10,11,12,13,17,22,23,24,26,27,28,29]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[2,3,4,5,9,14,15,16,18,19,20,21,25,30,31,32]

Table 7: Wyckoff site: 4g, site symmetry: 2mm.1'

No.	position	mapping
1	$[0, 0, z]$	[1,6,12,13,17,22,28,29]
2	$\left[0, 0, z + \frac{1}{2}\right]$	[2,3,15,16,18,19,31,32]

continued ...

Table 7

No.	position	mapping
3	$[0,\ 0,\ -z]$	[4,5,9,14,20,21,25,30]
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[7,8,10,11,23,24,26,27]

Table 8: Wyckoff site: 4h, site symmetry: 2mm.1'

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[1,6,12,13,17,22,28,29]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[2,3,15,16,18,19,31,32]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[4,5,9,14,20,21,25,30]
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[7,8,10,11,23,24,26,27]

Table 9: Wyckoff site: 4i, site symmetry: 2mm.1'

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,6,12,13,17,22,28,29]
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[2,3,15,16,18,19,31,32]
3	$\left[0,\ \frac{1}{2},\ -z\right]$	[4,5,9,14,20,21,25,30]
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[7,8,10,11,23,24,26,27]

Table 10: Wyckoff site: 4j, site symmetry: m2m.1'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4,13,14,17,20,29,30]
2	$\left[0,\ x,\ \frac{1}{2}\right]$	[2,7,11,16,18,23,27,32]
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	[3,8,10,15,19,24,26,31]
4	$[-x,\ 0,\ 0]$	[5,6,9,12,21,22,25,28]

Table 11: Wyckoff site: 4k, site symmetry: m2m.1'

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,4,13,14,17,20,29,30]
2	$\left[\frac{1}{2},\ x,\ 0\right]$	[2,7,11,16,18,23,27,32]
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	[3,8,10,15,19,24,26,31]
4	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[5,6,9,12,21,22,25,28]

Table 12: Wyckoff site: 4l, site symmetry: m2m.1'

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	[1,4,13,14,17,20,29,30]
2	$[0,\ x,\ 0]$	[2,7,11,16,18,23,27,32]
3	$[0,\ -x,\ 0]$	[3,8,10,15,19,24,26,31]

continued ...

Table 12

No.	position	mapping
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[5,6,9,12,21,22,25,28]`

Table 13: Wyckoff site: `4m`, site symmetry: `m2m.1'`

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	`[1,4,13,14,17,20,29,30]`
2	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[2,7,11,16,18,23,27,32]`
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[3,8,10,15,19,24,26,31]`
4	$\left[-x,\, \frac{1}{2},\, 0\right]$	`[5,6,9,12,21,22,25,28]`

Table 14: Wyckoff site: `8n`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	`[1,7,17,23]`
2	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[2,5,18,21]`
3	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[3,4,19,20]`
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	`[6,8,22,24]`
5	$\left[-x,\, -x,\, \frac{3}{4}\right]$	`[9,15,25,31]`
6	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[10,13,26,29]`
7	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[11,12,27,28]`
8	$\left[x,\, x,\, \frac{3}{4}\right]$	`[14,16,30,32]`

Table 15: Wyckoff site: `8o`, site symmetry: `.m.1'`

No.	position	mapping
1	$[0,\, y,\, z]$	`[1,12,17,28]`
2	$\left[-y,\, 0,\, z+\frac{1}{2}\right]$	`[2,15,18,31]`
3	$\left[y,\, 0,\, z+\frac{1}{2}\right]$	`[3,16,19,32]`
4	$[0,\, -y,\, -z]$	`[4,9,20,25]`
5	$[0,\, y,\, -z]$	`[5,14,21,30]`
6	$[0,\, -y,\, z]$	`[6,13,22,29]`
7	$\left[y,\, 0,\, \frac{1}{2}-z\right]$	`[7,10,23,26]`
8	$\left[-y,\, 0,\, \frac{1}{2}-z\right]$	`[8,11,24,27]`

Table 16: Wyckoff site: `8p`, site symmetry: `.m.1'`

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	`[1,12,17,28]`
2	$\left[-y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[2,15,18,31]`
3	$\left[y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[3,16,19,32]`
4	$\left[\frac{1}{2},\, -y,\, -z\right]$	`[4,9,20,25]`
5	$\left[\frac{1}{2},\, y,\, -z\right]$	`[5,14,21,30]`

continued ...

Table 16

No.	position	mapping
6	$\left[\frac{1}{2}, -y, z\right]$	[6,13,22,29]
7	$\left[y, \frac{1}{2}, \frac{1}{2} - z\right]$	[7,10,23,26]
8	$\left[-y, \frac{1}{2}, \frac{1}{2} - z\right]$	[8,11,24,27]

Table 17: Wyckoff site: 8q, site symmetry: m..1'

No.	position	mapping
1	$[x, y, 0]$	[1,14,17,30]
2	$\left[-y, x, \frac{1}{2}\right]$	[2,11,18,27]
3	$\left[y, -x, \frac{1}{2}\right]$	[3,10,19,26]
4	$[x, -y, 0]$	[4,13,20,29]
5	$[-x, y, 0]$	[5,12,21,28]
6	$[-x, -y, 0]$	[6,9,22,25]
7	$\left[y, x, \frac{1}{2}\right]$	[7,16,23,32]
8	$\left[-y, -x, \frac{1}{2}\right]$	[8,15,24,31]

Table 18: Wyckoff site: 16r, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,17]
2	$\left[-y, x, z + \frac{1}{2}\right]$	[2,18]
3	$\left[y, -x, z + \frac{1}{2}\right]$	[3,19]
4	$[x, -y, -z]$	[4,20]
5	$[-x, y, -z]$	[5,21]
6	$[-x, -y, z]$	[6,22]
7	$\left[y, x, \frac{1}{2} - z\right]$	[7,23]
8	$\left[-y, -x, \frac{1}{2} - z\right]$	[8,24]
9	$[-x, -y, -z]$	[9,25]
10	$\left[y, -x, \frac{1}{2} - z\right]$	[10,26]
11	$\left[-y, x, \frac{1}{2} - z\right]$	[11,27]
12	$[-x, y, z]$	[12,28]
13	$[x, -y, z]$	[13,29]
14	$[x, y, -z]$	[14,30]
15	$\left[-y, -x, z + \frac{1}{2}\right]$	[15,31]
16	$\left[y, x, z + \frac{1}{2}\right]$	[16,32]